Building Emotional Conversation Systems using Multi-task Seq2Seq Learning

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Abstract. This paper describes our system designed for the NLPCC 2017 shared task on emotional conversation generation. Our model adopts a multi-task Seq2Seq learning framework to capture the textual information of post sequence and generate responses for each type of emotions simultaneously. Evaluation results suggest that our model is competitive on emotional generation, which achieves 0.9658 on average emotion accuracy. We also observe the emotional interaction in human conversation, and try to explain it as empathy at the psychological level. Finally, our model achieves 325 on total score, 0.545 on average score and won the fourth place on total score.

Keywords: Conversation Generation, Emotions, Multi-task Seq2Seq

1 Introduction

The NLPCC 2017 emotional conversation generation shared task focuses on generating emotional coherent responses in Chinese. This task is challenging due to the difficulty of capturing emotional factors and the complex mechanism of human emotions. There have been prior works [1, 11, 14, 15]on emotional conversation, however, most of them are either rule-based or limited to small-scale data, not extensible to large-scale data.

Recently, large-scale conversation generation approaches[12, 16, 13] have been investigated, thanks to the advance of deep learning. Inspired by these end-to-end approaches, Zhou et al. proposed Emotional Chatting Machine(ECM)[20], a model in the encoder-decoder framework of large-scale sequence-to-sequence generation that can respond to users emotionally. However, ECM had to specify an emotion category to be generated, which need an external decision maker.

In this paper, we propose a multi-emotional conversation system(MECS) to generate emotional coherent responses. The model consists of two main components: (1) a multi-emotion response generator to generate responses for each type of emotions simultaneously, and (2) an output selector to choose the most appropriate response using intra-ranking and inter-ranking policies.

We evaluate our model at both context level and emotion level. The experimental results show that our model is competitive in emotional conversation

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generation task, which achieves 0.9658 on average emotion accuracy. We report on a detailed analysis of the evaluation results in the NLPCC 2017 shared task. The evaluation results indicate that the multi-task framework performs well on emotional conversation generation, although the single-layer GRU architecture is too simple to capture complex textual information.

In emotion interaction analysis, we also find that people tend to achieve the same emotional state in conversation. We try to explain this phenomenon as empathy at the psychological level.

The rest of the paper is arranged as follows. In Section 2 we discuss the detail of data preprocessing. In Section 3 we present MECS, our approach for emotional conversation generation task. In Section 4 we outline the experiments. We close with our conclusions in Section 5, and a discussion of future work.

2 Preprocessing

2.1 Data Cleaning

There are some specific symbols, excess punctuations and Cantonese text in raw text. We clean the text by removing the specific symbols and excess punctuations. In order to remove Cantonese text from the raw data, we crawl some Cantonese text and Chinese text as training data from Weibo, and construct a classify model using convolutional neural network introduced by Kim[8]. After data preprocessing, about 85.9% of post-response pairs have been retained, as shown in Table 1.

Table 1. Data distribution of clean data.

Categories of Responses	Original Data	Clean Data
Other	198,069	172,197
Like	200,001	169,371
Sadness	181,252	158,657
Disgust	200,001	176,207
Anger	139,883	115,740
Happiness	200,001	168,821
Total	1,119,207	960,993

2.2 Response Expansion

We expand the data for training a multi-task emotional conversation model, since the format of the raw data is one-to-one: each pair contains one post and only one response, together with their emotional information.

The training data are firstly grouped according to the emotion categories of responses. Each post-response pair is expanded to a tuple with one post and six

emotional responses, corresponding to different emotion types respectively. The original emotional response is retained, while the missing responses are filled by searching the most appropriate responses from the corresponding emotional groups using a retrieve-based method. The retrieve-based approach is introduced as follows:

We train a LDA topic model using gensim[1], and calculate text similarity using both topic similarity and cosine similarity:

$$F_{sim}(p_1, p_2) = \lambda S_{topic}(p_1, p_2) + (1 - \lambda) S_{cosine}(p_1, p_2) \tag{1}$$

where λ is set to 0.5. S_{topic} is topic similarity calculated by the LDA model and S_{cosine} is cosine similarity defined as:

$$S_{cosine}(p_1, p_2) = \frac{\overrightarrow{v_1} \cdot \overrightarrow{v_2}}{\|\overrightarrow{v_1}\| \, \|\overrightarrow{v_2}\|} \tag{2}$$

where p_1 and p_2 are post text, corresponding to word vectors $\overrightarrow{v_1}$ and $\overrightarrow{v_2}$ respectively.

Formula (1) calculates the similarity between two posts at both text-level and topic-level. Yet the retrieve-based model picks out the most similar pair evaluated by Formula (1), and returns the respective response as result.

To measure the appropriateness of expanded training pairs, we employ human judgements to evaluate a random sample of 500 items. Each item is assigned to 3 judges, who are asked to decide whether the responses are topic coherent. Results for human evaluation show that about 40.93% of expanded responses is acceptable.

3 System Description

3.1 Multi-emotional Conversation Generation

Inspired by Luong et al.[10], we propose a multi-emotional conversation system. Given a new post, our model is to generate six responses simultaneously, corresponding to each kind of emotions respectively, which are {other, like, sadness, disgust, anger, happiness}, as shown in Figure 1. The encoder and decoders of the multi-task Seq2Seq model are implemented with GRU[4], since GRU performs comparable to LSTM[5] but has less parameters and easier to train.

One-to-many Architecture In order to generate responses that are coherent with the target emotion categories, we adopt a one-to-many Seq2Seq architecture. Each decoder is independent of others, but shares the same encoder.

Given a post sequence $\mathbf{X} = (x_1, x_2, \cdots, x_n)$, the encoder converts $\mathbf{X}$ to its hidden representations $\mathbf{h} = (h_1, h_2, \cdots, h_n)$ using GRU, briefly defined as:

$$h_t = \mathbf{GRU}(h_{t-1}, x_t) \tag{3}$$

[1] https://radimrehurek.com/gensim/

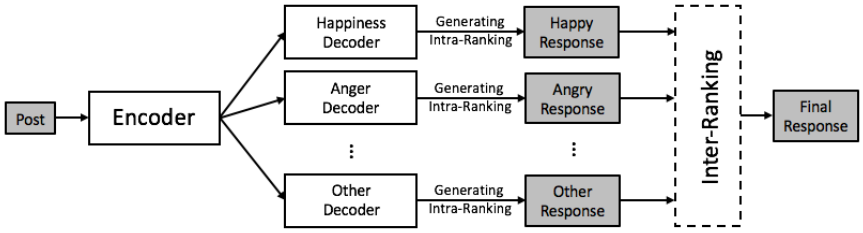


Fig. 1. The general framework and dataflow of MECS.

Details of GRU is introduced by Cho et al.[4].

Each decoder takes the embedding of a previously decoded word y^i_{t-1} and a context vector c^i_t, and updates its state s^i_t using another GRU:

$$s^i_t = \mathbf{GRU}(s^i_{t-1}, [c^i_t; e(y^i_{t-1})]) \tag{4}$$

where i is the emotional decoder index, $e(y^i_{t-1})$ is the corresponding word vector of y^i_{t-1}, and $[c^i_t; e(y^i_{t-1})]$ is the concatenation of the two vectors. The attention mechanism[2] is applied to learn context vector c^i_t, which depends on the previous state s^i_{t-1} of the decoder:

$$c^i_t = \sum_{t=1}^{T} \alpha^i_{kt} h_t \tag{5}$$

$$\alpha^i_{kt} = \frac{exp(e^i_{kt})}{\sum_{j=1}^{T} exp(e^i_{kj})} \tag{6}$$

$$e^i_{kt} = (v^i_a)^\top tanh(W^i_a s^i_{k-1} + U^i_a h_k) \tag{7}$$

where α^i_{kt} is the weight of hidden layer representation h_t of the post sequence for the ith decoder. It is noteworthy that the hidden representations $\mathbf{h}$ of post sequence are shared between each emotional decoder.

Finally, the output probability distribution o^i_t of the next token y^i_t is computed as:

$$y^i_t \sim o^i_t = P(y^i_t | y^i_{t-1}, \cdots, y^i_1, c^i_t) \tag{8}$$

$$= softmax(W^i_o s^i_t) \tag{9}$$

and the next token y^i_t is generated by sampling from the distribution o^i_t.

Training The objective function is based on the cross entropy error between the predicted token distribution o^i_t and the gold distribution p^i_t in the training corpus, defined as:

$$L(\theta) = -\sum_t {p^i_t}^\top log(o^i_t) \tag{10}$$

Since each task in MECS is similar, we adopt an alternating training approach as introduced by Dong et al.[6].

3.2 Ranking Policy

Though the multi-emotional conversation system could generate responses for all emotion types, however, a real conversation system should return only one response, instead of six. Therefore, we propose intra-ranking and inter-ranking policies to select the best response as output.

Intra-ranking For each decoder, we adopt beam-search to over-generate responses. These responses are ranked using:

$$p(\mathbf{Y}|\mathbf{X}) = \prod_{t=1}^{N_y} p(y_t|x_1, x_2, \cdots, x_T, y_1, y_2, \cdots, y_{t-1}) \tag{11}$$

$$= \prod_{t=1}^{N_y} \frac{exp(f(s_t, e_{y_t}))}{\sum_{y'} exp(f(s_t, e_{y'}))} \tag{12}$$

where $f(s_t, e_{y_t})$ denotes the corresponding output on the last projection layer for word y_t, and N_y denotes the length of response sequence $\mathbf{Y}$.Ultimately, we select the sentence with highest score as the output of this decoder.

Inter-ranking After responses for different emotion types have been generated, an inter-ranking policy is adopted to select the most appropriate response as the final output. In MECS, however, we rank the results among emotional decoders directly based on the score calculated by Formula (11). This naive policy tends to choose a fluent expression without considering the emotion type of input text. Therefore, it is not capable to simulate the emotion interaction in human conversation, as discussed in Chapter 4.2.

4 Experiments

4.1 Model Details

We adopt single-layer GRU for encoding and decoding, each of which consists of a different set of parameters. Each GRU layer consists of 128 hidden neurons, and the dimensionality of word embedding is also set to 128. The model is trained on a single GTX1060 GPU for less than 3 days. Other training details are given as below.

- GRU parameters and embeddings are initialized from a uniform distribution in [0,1).
- Stochastic gradient decent is implemented and learning rate is set to 0.5, with a decay rate of 0.99.
- Batch size is set to 128. Epoch size is set to 20.
- Beam-search is adopted with beam size of 5, and buckets are applied to generate response with appropriate length. Buckets are set to [(5,10), (10, 15), (20, 25), (40, 50)].

4.2 Experimental Results and Analysis

Evaluating a response generation model is still an open problem. Following [17] we employ perplexity as an evaluation metric at the context level, since BLEU is not suitable for measuring conversation generation, as argued in [9]. To evaluate the model at the emotion level, we adopt emotion accuracy as the agreement between the expected emotion category and the predicted emotion category of the response generated by the corresponding emotional decoder, using the same Bi-LSTM emotion classifier mentioned in the homepage[2] of the shared task. The results are shown in Table 2.

Table 2. Quantitative evaluation in terms of perplexity and emotion accuracy.

Categories of Responses	Perplexity	Emotion Accuracy
Other	383.86	0.9467
Like	79.14	0.9895
Sadness	77.26	0.9886
Disgust	318.73	0.9171
Anger	103.33	0.9533
Happiness	54.36	0.9996
Average	169.45	0.9658

As can be seen, our model performs well in emotion accuracy, but the perplexity of the decoders varies widely, since each decoder is trained on definitely different pairs. Thanks to the multi-task framework, each decoder is trained to respond in a specific emotional state, tends not to generate response with ambiguous emotion. Furthermore, tokens that express other emotions might not appear in the training pairs for the specific decoder. For example, the words "resentful" tends not to appear in a happy response. Hence, we can even customize the dictionary for each decoder separately, to reduce model parameters and improve emotion accuracy.

However, it is worth noting that the accuracy rate here is falsely high more or less. Since we use only single-layer GRU for encoding and decoding, the model tends to generate dull responses. For example, the word "haha", which is also a significant identifier token for the emotion classifier, appears almost in every result in "happiness" category.

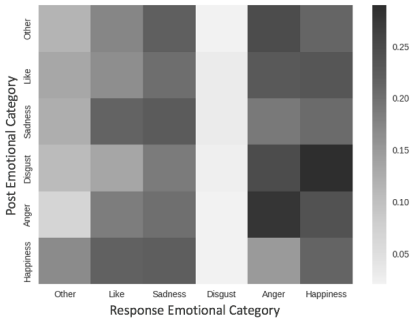
4.3 Emotion Interaction Analysis

Moreover, we evaluate the inter-ranking mechanism by visualizing the emotion interaction. Emotion interaction pattern(EIP) is defined as $< e_p, e_r >$, as introduced in [20], is the pair of emotion categories of a post and its response. The

[2] http://www.aihuang.org/p/challenge.html

(a) EIP of Training Data

(b) EIP of Evaluation Data



(c) EIP of Inter-ranking Results

Fig. 2. Visualization of emotion interaction.

value of an EIP is the conditional probability $P(e_r|e_p) = P(e_r, e_p)/P(e_p)$. We visualize the emotion interaction pattern as Figure 2.

Unlike the analysis of [20], we observed some interesting features in training data, as shown in Figure 2.(a). Obviously, the color of the blocks on the diagonal is much darker, which means that people tend to response in the same emotional state in a conversation. We believe that this phenomenon can be explained by empathy. Empathy is the capacity to understand what another person is experiencing from within the other person's frame of reference, as defined in [3]. The same phenomenon is also observed in the evaluation data, as shown in Figure 2.(b), although the evaluation data does not contain responses in "other" category.

Unfortunately, as shown in Figure 2.(c), the inter-ranking policy is not capable to simulate the empathy phenomenon. As discussed in Chapter 3.2, the inter-ranking mechanism tends to select sentences generated by the decoder which has lower perplexity, for example, the decoder of "happiness" category. On the other

hand, responses generated by a decoder with higher perplexity, such as the decoder of "other" or "disgust" category, are difficult to be selected. Therefore, the results of this ranking mechanism might be focused, although it can guarantee the fluency of the response.

4.4 Shared Task Evaluation Results

Results of our model and other top competitors in the shared task are shown in Table 3. Our model achieves 325 on total score and won the fourth place on this indicator. According to the summary report, the number of results submitted by each team is very different, so the total score and the average score can reflect the effect of the model to a certain extent. The table below lists only the top-ranked teams which completely submit the results.

Table 3. Human evaluation results in NLPCC 2017 shared task.

Model	0 Points Terms	1 Point Terms	2 Points Terms	Total Terms	Total Score	Average Score
samsung-1	267	82	247	596	276	0.966443
Babbling-2	305	19	271	595	561	0.942857
samsung-2	272	93	231	596	555	0.931208
Ours	**428**	**11**	**157**	**596**	**325**	**0.545302**
Babbling-1	424	41	131	596	303	0.508389
gry-1	449	31	116	596	263	0.441275

We also conducted a detailed analysis of each emotional decoder. The human evaluation results and analysis are as follows:

Table 4. Human evaluation results among emotional categories for MECS.

Emotion Categories	0 Points Terms	1 Point Terms	2 Points Terms	Total Terms	Total Score	Average Score
Like	100	0	33	133	66	0.496241
Sadness	82	0	37	119	74	0.621849
Disgust	133	6	5	144	16	**0.111111**
Angry	59	2	10	71	22	0.309859
Happiness	54	3	72	129	147	**1.139535**

As can be seen, our model generates a large amount of 0 points terms, which means the generation result is not coherent or not fluent. Due to the poor capability of single-layer GRU to capture semantics features, our model is not sufficient to produce high-quality responses. The evaluation results are consistent with

the analysis in Table 2: the "happiness" decoder with lowest perplexity generates responses with highest quality, and decoders with high perplexity (such as "disgust" decoder) performs poorly.

On the other hand, our model generates few 1 point terms, which means the generation result is coherent and fluent but not emotionally consistent. As discussed in Chapter 4.2, our model achieves a high emotional accuracy. The human evaluation result here is also consistent with our experimental result, which shows that our model performs well in the emotional generation task.

5 Conclusions and Future Works

In this paper, an emotional conversation system based on multi-task Seq2Seq framework is proposed. Although this model behaved moderately in NLPCC 2017 shared task, the evaluation of further experiments shows that multi-task framework could behave competitively on emotional generation.

Our approach leaves a lot of room for extensions. First of all, the single-layer GRU architecture is too simple to capture the complex information of input text, and is not capable enough to generate informative and interesting responses. In addition, with our naive inter-ranking policy, this model cannot simulate the empathy phenomenon in human conversation.

For future work, we plan to enhance the ability of our model to generate topic coherence responses, as discussed in [18, 21, 19, 7]. We also consider designing an appropriate ranking policy to capture the emotion interaction and make the conversation system more human.

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